What Do Vision Transformers Read? Executable Attention Reading Sets under Centre-Bias Control

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Abstract

Attention maps are the most common way to look inside Vision Transformers (ViTs), but they are hard to verify and easily confused with the centre bias of photographs. We read, from unmodified pretrained ViTs, which tokens each image region attends to: following the fast-weight view of attention and the top-k reading rule of MiTA, each of 16 regions keeps the 16 keys that its mean query scores highest. Because these reading sets can be executed, we can check that they are what the model uses: restricting all twelve layers to them keeps 72% of DeiT-S top-1 predictions, against 9% for random sets of the same size and 0.1% with no attention at all. We then ask what three attention readouts—the top-k sets, dense attention received, and CLS attention—reveal about DeiT-S, DINO ViT-S and DeiT-B under tests that control for centre bias. Attention goes to objects beyond the centre prior, follows objects moved off-centre, and agrees with human importance maps image by image; the models themselves rely more on animal heads than bodies. Layer by layer, reading is most human-like in mid-to-late layers, and the last layer of the supervised ViTs, but not of DINO, drifts off the object. Supervised ViTs route up to 61% of mid-to-late reading to about 3% of high-norm tokens, which DINO lacks, yet this does not explain the drift. As saliency maps, all attention readouts remain clearly less faithful than gradient-based GAE, and the top-k step does not improve the map over dense attention.

1 Introduction

Attention maps are cheap and look plausible, which makes them the default view into ViTs [Abnar and Zuidema, 2020, Caron et al., 2021]. Two problems limit what they tell us. A map is a picture, not a computation: nothing guarantees that the model relies on the highlighted tokens [Jain and Wallace, 2019, Wiegreffe and Pinter, 2019]. And objects in photographs sit near the centre, so a map that always highlights the centre already scores well; a fixed centred Gaussian correlates at 0.41 with ClickMe human importance maps [Fel et al., 2022], more than our probe does on any of three models.

One attention layer computes the same function as a two-layer network whose first-layer weights are the keys, whose activation is the softmax and whose second-layer weights are the values [Schlag et al., 2021, Wen et al., 2026]. Each token is then a hidden neuron tied to an image position. MiTA [Wen et al., 2026] trains models that read only the top-k such neurons per image region; we instead *read* these sets from unmodified pretrained ViTs and *execute* the model through them, so the reading is checked by the model's own output. Around this probe we build tests that remove centre bias and compare three attention readouts with gradient-based explanations on three models and five datasets.

Contributions. (i) Executable reading sets that keep 62–83% of predictions when every layer reads only them, against 4–30% for random sets (Sec. 2). (ii) A centre-controlled protocol: same-distance-ring enrichment, image-specific human alignment, centre-matched and part-wise token removal

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(Sec. 3). (iii) Findings that hold for three readouts: attention and the model's reliance go to objects and animal heads beyond the centre; reading is most human-like in mid-to-late layers; the last layer of supervised ViTs drifts off the object, DINO's does not; high-norm tokens absorb much mid-to-late reading but do not explain the drift (Sec. 4.3). (iv) A negative result: attention readouts trail GAE on deletion faithfulness, and top-k selection does not improve the map over dense attention (Sec. 4.2). All predictions and pass criteria were registered before the runs (Appendix C).

Related work. Rollout [Abnar and Zuidema, 2020] propagates attention through layers; GAE and related methods combine attention with gradients [Chefer et al., 2021b,a]; AttnLRP, TokenTM and LibraGrad refine relevance propagation [Achtibat et al., 2024, Wu et al., 2024b, Mehri et al., 2025]. We evaluate with deletion curves [Petsiuk et al., 2018] summarised by SRG [Blücher et al., 2024] under token removal [Jain et al., 2022, Mehri et al., 2025], SaCo [Wu et al., 2024a], the randomisation sanity check [Adebayo et al., 2018] and the Harmonization protocol [Fel et al., 2022]. Top-k attention [Gupta et al., 2021, Wen et al., 2026] targets efficiency, and "attention received" scores tokens for pruning [Goyal et al., 2020, Kim et al., 2022, Wang et al., 2024]; our dense readout is this score, used here as an explanation. High-norm artefact tokens [Darcet et al., 2024] appear in supervised ViTs but not in the original DINO, which our measurements reproduce.

2 Method

Reading sets. For a layer and head with queries q_i and keys k_j (j over 197 tokens), we pool the 14×14 patch grid into $m = 16$ regions (4×4), take each region's mean query $\bar{q}_r$, and keep $S_r = \text{top-}\, k_j\, \bar{q}_r^\top k_j$ with $k = 16$. Every patch query reads only the set of its region; CLS reads its own top-k set. Sets come from one ordinary forward pass and do not change the model. To check them (*execution test*), we run the model with selected layers restricted to the sets and measure top-1 agreement with the unmodified model, against random sets of the same size.

Three readouts. For layers $\ell \in \mathcal{L}$, we score patch j by the attention it receives:

$$s_j = \sum_{\ell \in \mathcal{L}} \sum_h \sum_i \frac{\exp(q_i^\top k_j / \sqrt{d})\, [j \in S_{r(i)}]}{\sum_{j' \in S_{r(i)}} \exp(q_i^\top k_{j'} / \sqrt{d})}. \tag{1}$$

The *probe* uses the reading sets as written; *dense attention received* drops the indicator (all 197 tokens, the token-pruning score [Goyal et al., 2020]); *CLS attention* keeps only the CLS query. Maps use layers 9–12 unless a single layer is stated; none needs gradients or a class label, and the probe costs 1.1 forward passes (GAE 2.6, Integrated Gradients 46).

3 Evaluation protocol

Models and data. DeiT-S/16 and DeiT-B/16 (supervised) [Touvron et al., 2021] and DINO ViT-S/16 with its official linear head [Caron et al., 2021]. ImageNet-S val (12,419 images with masks) [Gao et al., 2022], Hard ImageNet (750; small, off-centre objects) [Moayeri et al., 2022], Oxford Pet [Parkhi et al., 2012], ClickMe val (17,408 human importance maps) [Fel et al., 2022] and PartImageNet (24,080 part masks; images from the ImageNet *training* set) [He et al., 2022].

Measures. *Faithfulness*: patches are removed from every layer by dropping their tokens; SRG is the area between the least- and most-relevant-first deletion curves of top-1 agreement (0 for random scores). *Same-ring enrichment*: among patches at the same distance from the centre, the object share of a map's top patches minus the ring's object share; a centre prior scores 0. *Image-specific human alignment*: Spearman correlation with the image's own ClickMe map minus that with another image's map; a fixed prior scores 0. *Parts*: head enrichment is a map's mass share on the head divided by the head's area share; the model's own reliance is the drop in log-probability when four patches of one part are removed. Intervals are 95% paired bootstrap intervals over images (1,000 resamples).

4 Results

4.1 The reading sets are what the model computes with

The test is the whole network: restricting all twelve layers keeps 72% (DeiT-S), 62% (DINO) and 83% (DeiT-B) at $k = 16$, against 9%, 4% and 30% for random sets and 0.1–0.2% with no attention at

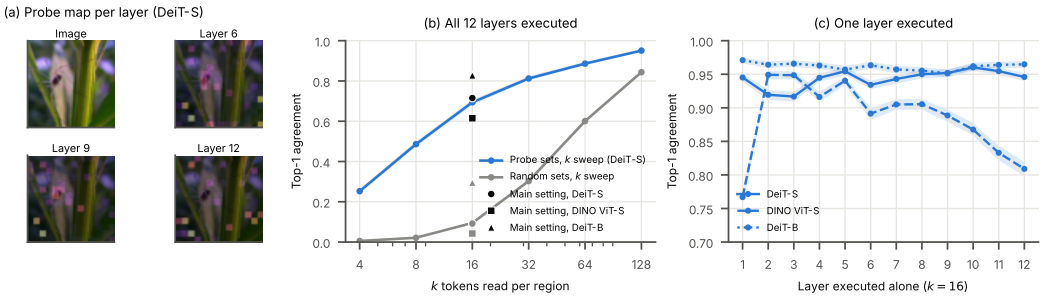


Figure 1: **Reading sets are executable.** (a) Single-layer probe maps of a random ImageNet-S image. (b) Top-1 agreement when all 12 layers read only the reading sets; the sweep uses a first-generation setting (CLS shares its region's set), markers the main setting at $k = 16$ for each model (grey: random sets). (c) One layer executed at a time (5,000 ImageNet images; bands: 95% intervals).

Table 1: **Attention readouts and gradient methods under the same tests** (S: DeiT-S, D: DINO ViT-S, B: DeiT-B). SRG and enrichment on ImageNet-S, human alignment on ClickMe, head enrichment on PartImageNet animals (1 = no preference). †Uses gradients and the predicted class. Bold: best attention readout. Dense attention comes from a second run on another GPU; the probe re-run there differs from these values by at most 0.001 SRG.

	SRG ↑			Same-ring enrichment ↑			ClickMe image-specific ρ ↑			Head enrichment		
Method	S	D	B	S	D	B	S	D	B	S	D	B
Probe (top-k sets)	0.201	0.235	0.150	0.114	0.158	0.067	0.135	0.199	0.092	**1.54**	**1.82**	**1.30**
Dense attention received	**0.242**	0.285	**0.169**	**0.140**	0.199	**0.080**	**0.156**	0.208	**0.107**	1.25	1.26	1.17
CLS attention	0.182	**0.309**	0.110	0.089	**0.238**	0.047	0.083	**0.217**	0.032	1.06	1.34	0.88
Rollout	0.146	0.172	0.030	0.110	0.144	0.045	0.141	0.171	0.050	1.09	1.04	1.01
GAE†	0.343	0.418	0.308	0.204	0.221	0.196	0.234	0.234	0.213	1.28	1.35	1.17
Grad-CAM†	0.341	0.316	0.274	0.186	0.120	0.170	0.208	0.145	0.172	1.10	1.18	1.04
Integrated Gradients†	0.037	0.230	-0.001	0.008	0.152	0.004	0.008	0.195	-0.010	1.01	1.23	0.96
Centre Gaussian (reference)	0.109	0.142	0.100	0.000	0.000	0.000	0.000	0.000	0.000	1.00	0.99	1.00

all. Single layers are no such test: one layer restricted to its sets keeps 92–96% of DeiT-S predictions (Fig. 1c), but deleting that layer's attention outright already keeps 93–97% (69% at layer 1), so they only show which layers are harder to replace. Reading concentrates with depth: in DeiT-S the union of all sets covers 75% of patches at layer 3 but 12% at layer 12. Attention reading can therefore be treated as a measured quantity rather than a picture.

4.2 Attention readouts as saliency maps

Table 1 compares the readouts with GAE [Chefer et al., 2021a], Grad-CAM [Selvaraju et al., 2017] and Integrated Gradients [Sundararajan et al., 2017]. (1) A model-free centre Gaussian already reaches SRG 0.10–0.14 and beats Rollout on DeiT-B, so centre bias matters for faithfulness metrics too. (2) Among attention readouts, dense attention received is the most faithful on DeiT-S and DeiT-B and CLS attention on DINO, whose classifier reads the CLS token of the last four blocks; even the best of them trails GAE by 0.10–0.14 SRG. The same ordering holds on Hard ImageNet, Pet and an ImageNet subset (Appendix B). (3) The top-k probe and dense attention are nearly the same map (median per-image Spearman 0.90 on ImageNet-S). Dense attention is equal or better on faithfulness (higher in all 10 dataset–model pairs, by 0.006–0.050, significantly in 9), object focus and human alignment; the probe's stronger head preference (1.54 vs. 1.25 on DeiT-S) comes from truncation and does not make it better at naming the part the model relies on most (46.7% vs. 46.4%, chance 40%). The attention readouts we tested (probe, CLS attention, Rollout) pass the randomisation sanity check (median Spearman after re-initialising all blocks: probe 0.08/0.32/0.10), while Integrated Gradients fails on all three models (0.53–0.61). We therefore use dense attention as the map and the top-k sets as its verification.

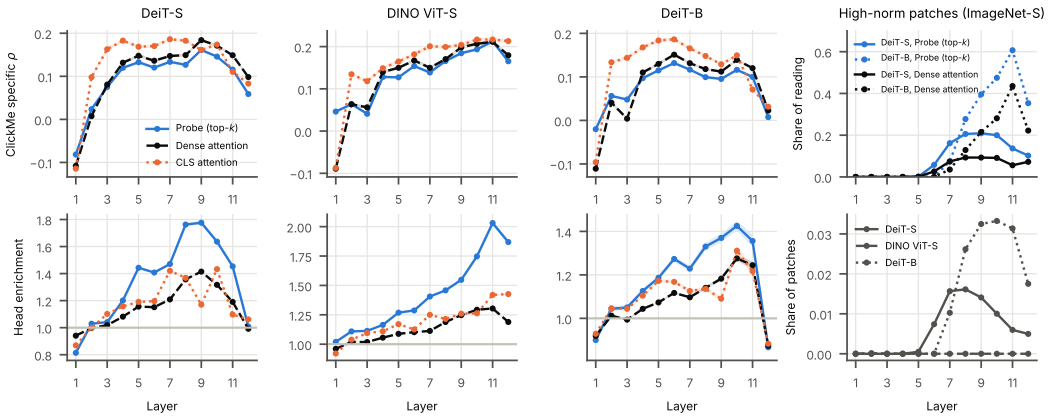


Figure 2: **Reading layer by layer.** Columns 1–3: image-specific human alignment (ClickMe, top) and head enrichment (PartImageNet, bottom) of single-layer maps for three readouts. Column 4: share of single-layer reading on high-norm patches (block-input norm $> 3\times$ the image's median) and share of such patches on ImageNet-S; DINO has none.

4.3 What the models read

Beyond the centre. All readouts have positive same-ring enrichment (Table 1): reading favours objects even among patches equally far from the centre. When an object is shifted 48 pixels towards a corner, the probe's object mass drops less than a centre prior's (by 0.021, 0.053, 0.010 for DeiT-S, DINO, DeiT-B). DeiT-S and DINO depend on objects, not positions: removing object tokens from the reading sets hurts agreement more than removing as many tokens at matched distances from the centre, and 63–79% of the object effect survives this matching on ImageNet-S and Hard ImageNet.

Heads. Removing four head patches lowers the log-probability of the predicted class more than removing four body patches of the same image ($+0.017$ [0.014, 0.020] DeiT-S, $+0.021$ DINO, $+0.008$ DeiT-B), 2–3.4 times as much for quadrupeds and bipeds; the probe, dense attention and GAE favour heads on all three models (enrichment > 1).

Layers. Human alignment and head preference rise through the network and peak in mid-to-late layers (DeiT-S layer 9, DINO layer 11, DeiT-B layers 6–10; Fig. 2). In the supervised models the last layer drops sharply (dense attention, layer 12 vs. the best of layers 8–11: ClickMe -0.086 DeiT-S, -0.116 DeiT-B; head enrichment -0.42, -0.40), whereas DINO drops little (-0.032; CLS attention -0.004). The pattern is the same for all three readouts (correlation of layer curves with the probe's: 0.77–0.99), so it is a property of the models' attention, not of the probe.

High-norm tokens. Supervised ViTs route much mid-to-late reading to a few high-norm patches: in DeiT-B layer 11, 3% of patches receive 61% of the probe's and 44% of dense attention. DINO has no such patches, as reported by Darcet et al. [2024]. Yet this does not explain the last-layer drift: the probe's high-norm share is lower at layer 12 than at layers 10–11 in both supervised models (our registered prediction failed for DeiT-S), so why the last layer leaves the object remains open.

5 Discussion and limitations

Executable reading sets make attention a checkable measurement, and centre-controlled tests show that what ViTs read goes beyond where photographers put objects. Attention readouts are class-agnostic and less faithful than gradient-based, class-specific GAE; we did not test whether class specificity explains the gap. Part results use ImageNet training images, which the models have seen. Deep tokens drift from their patches: a linear probe recovers a layer-12 token's exact patch only 15–25% of the time (chance 0.5%), so deep maps localise to about 1–3 patches. Each configuration ran once; intervals reflect image sampling only. *Broader impact.* Plausible-looking explanations can create unjustified trust, which is why we report centre references and negative results; the datasets are public benchmarks with known demographic and geographic biases.

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A Reproducibility

Pretrained weights are pinned by revision and checksum (timm DeiT-S/B, official DINO ViT-S/16 and its linear head). All runs use float32 without autocast and deterministic algorithms, on one NVIDIA A100 (main runs) or L4 (dense-attention runs). Each run stores its configuration, log, git commit and result tables; tables and figures are generated from those files by scripts (`scripts/paper_assets.py`). Faithfulness uses all ImageNet-S and Hard ImageNet images for DeiT-S and DINO and a fixed random 2,000-image subset (seed 0) for DeiT-B, Pet and the ImageNet subset. Removing a patch drops its token after the positional embedding, which equals masking it in every layer's attention (unit-tested to 2×10^{-5} in the logits).

B Additional results

C Registered predictions

Every prediction and its pass criterion was written down before the runs. Of the 29 directional predictions in our ledger, 19 held, 7 held partly and 3 failed. Failed: the probe is *not* as faithful as GAE (all 10 pairs); the centre prior's head enrichment is about 1, not below 1; and, in a direction favourable to us, the DeiT-S and DeiT-B probes do show object preference beyond the centre (ring enrichment $+0.114$, $+0.067$) where we expected none. Partly held: the probe beats CLS attention on DeiT-S and DeiT-B but not on DINO; Integrated Gradients fails the sanity check and does not beat random scores on DeiT-B; "dense attention is comparable to the probe" held for SRG but not fully for object focus and human alignment, where dense attention is better; the high-norm share rises at the last layer for DeiT-B but not DeiT-S.

Table 2: SRG on every dataset and model (S: DeiT-S, D: DINO, B: DeiT-B).

Method	ImageNet-S S	ImageNet-S D	ImageNet-S B	Hard ImageNet S	Hard ImageNet D	Hard ImageNet B	Pet S	Pet D	ImageNet S	ImageNet D
Probe (top-k sets)	0.201	0.235	0.150	0.177	0.192	0.136	0.252	0.328	0.205	0.235
Dense attention received	0.242	0.285	0.169	0.198	0.233	0.142	0.278	0.375	0.241	0.277
CLS attention	0.182	0.309	0.110	0.135	0.242	0.100	0.179	0.387	0.179	0.301
Rollout	0.146	0.172	0.030	0.134	0.126	0.025	0.210	0.271	0.149	0.170
GAE†	0.343	0.418	0.308	0.325	0.397	0.316	0.416	0.480	0.361	0.425
Grad-CAM†	0.341	0.316	0.274	0.315	0.328	0.279	0.360	0.351	0.345	0.335
Integrated Gradients†	0.037	0.230	-0.001	0.034	0.183	0.012	0.060	0.298	0.047	0.221
Centre Gaussian (reference)	0.109	0.142	0.100	0.064	0.115	0.061	0.160	0.202	0.114	0.143
Random (reference)	-0.001	-0.001	-0.003	0.002	-0.007	-0.003	0.010	0.001	0.001	-0.000